SYNTHETIC DEMO – NOT RESULTS

A. Ablations on Test B (held-out cycling conditions)

Arm	Hyp.	What changes relative to A1	Param.	ΔRMSE (pp) [95% BCa]	Ratio [95% int.]	Tier
A9	H1a	no residual, no law ($\lambda = 0$): data-only F with the same z_c, inputs and window inversion	50945	0.86 [0.57, 1.24]	1.84 [1.31, 2.27]	S
A3	H1b	cyclic stress map: log-linear Arrhenius / power-law factors	410	0.29 [0.14, 0.47]	1.30 [1.06, 1.52]	desc.
A4	H1b	cyclic stress map: additive GAM (KAN without hidden layer)	2623	0.12 [−0.10, 0.33]	1.13 [0.95, 1.31]	desc.
A5	H1b	cyclic stress map: GAM with three tensor-product terms	3312	−0.00 [−0.18, 0.14]	1.15 [0.97, 1.24]	N
A7	H1b	cyclic stress map: tanh MLP, parameters capped at the largest A1 map	2790	0.23 [0.07, 0.42]	1.22 [0.99, 1.37]	desc.
A2	H2	exact integration of the same law, no F and no residual (hard ODE)	188	0.12 [−0.02, 0.27]	1.13 [0.99, 1.30]	desc.
A13	X1	trained on cycling cells only (no storage cells)	2621	−0.05 [−0.20, 0.07]	0.97 [0.82, 1.07]	desc.
A10	H6	dataset authors' semi-empirical superposition with per-cell factor	11	0.38 [0.20, 0.57]	1.47 [1.08, 1.76]	S

B. Controls

Control	Statistic	Observed	Required / outcome
Negative: stress vectors permuted among conditions (A1, A9, R2 re-run)	mean L, A9 vs A1	0.004, two-sided $p = 0.62$	$< \ln 1.02$ and $p > 0.05$: pass
	median L, A1 vs R2	−0.020	no win beyond margin: pass
Positive P1: well-specified synthetic G0	nRMSE of recovered surface	–	≤ 0.15: pass
Positive P2: storage cells	calendar rate against T	monotone at every SOC in the hull	pass
Positive P3: PINN-KAN against persistence	median L	1.38	> 0: pass

Table 1: SYNTHETIC DEMO – NOT RESULTS. Ablations and controls. Part A: each row changes one component of PINN-KAN (A1). ΔRMSE = comparator minus A1, mean over held-out conditions with its 95 % BCa interval; ratio = median over conditions of comparator/A1 error with its exact 95 % order-statistic interval (above 1: A1 better); tier from the preregistered sign test (S superior, N non-inferior, – neither; n.t. gated; desc. descriptive). Pool arms A3–A7 are descriptive, Holm over the four. Part B: controls of PREREG_v2 section 10; $L = \ln$(error of comparator / error of A1) per condition.